

ST STEPHEN, NB, Canada

WMO: 716070

Lat: 45.2103N

Lon: 67.2528W

Elev: 26

StdP: 101.01

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-20.1	-28.4	0.3	-20.9	-25.8	0.4	-18.7	8.8	-1.3	7.9	-4.4	1.1	290	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.8	20.9	28.0	20.1	26.3	19.0	22.4	27.7	21.4	26.3	20.3	24.6	3.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.6	15.3	25.0	19.7	14.4	23.7	18.7	13.6	22.6	65.9	27.6	62.2	26.3	58.6	24.7	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.7	-27.1	33.1	3.4	1.3	-29.6	34.0	-31.6	34.8	-33.5	35.6	-36.0	36.5		
			-27.5	23.9	2.9	1.0	-29.6	24.6	-31.3	25.2	-32.9	25.7	-35.1	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-7.7	-6.3	-1.4	5.0	10.9	15.8	19.3	18.7	14.5	8.3	2.5	-3.6
	DBStd	10.32	6.61	6.09	5.25	3.70	3.36	3.33	2.64	2.75	3.84	3.99	4.79	5.65
	HDD10.0	2280	549	457	353	154	27	1	0	0	5	81	229	423
	HDD18.3	4470	808	690	611	399	230	89	20	29	126	311	476	681
	CDD10.0	968	0	0	1	5	57	176	289	268	141	28	4	0
	CDD18.3	116	0	0	0	0	1	14	51	39	11	0	0	0
	CDH23.3	1272	0	0	2	2	43	209	511	385	116	4	0	0
	CDH26.7	307	0	0	0	1	8	54	124	89	31	0	0	0
Wind	WSAvg	2.2	2.4	2.5	2.7	2.6	2.3	1.9	1.7	1.6	1.7	2.0	2.2	2.3
Precipitation	PrecAvg	1262	99	87	110	96	100	99	93	83	103	136	121	135
	PrecMax	1802	206	188	191	175	201	194	281	192	260	310	194	300
	PrecMin	775	41	23	32	24	26	45	24	9	44	49	60	55
	PrecStd	236	37	36	40	38	43	42	56	48	47	66	42	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	10.5	16.5	21.0	27.4	31.2	32.0	31.5	30.7	23.4	17.0	13.1
		MCWB	10.9	8.1	9.8	12.0	17.5	20.9	22.1	21.7	22.1	16.8	14.7	12.2
	2%	DB	8.3	6.8	10.5	17.1	24.0	28.2	29.6	29.0	26.5	19.7	14.7	9.7
		MCWB	7.4	4.3	6.0	9.8	15.2	19.7	21.1	20.8	19.4	14.8	13.0	8.8
	5%	DB	4.7	4.5	8.2	14.4	21.1	25.7	27.9	27.1	23.8	17.4	12.8	7.1
		MCWB	3.5	2.5	4.9	8.7	13.6	18.0	20.2	20.0	18.1	14.2	10.9	5.9
	10%	DB	1.7	2.7	6.0	12.2	18.7	23.5	26.2	25.4	21.6	15.6	10.5	4.6
		MCWB	0.7	1.1	3.2	7.6	12.2	16.6	19.3	18.8	16.7	12.9	8.8	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	8.7	11.4	14.3	19.2	22.7	23.7	23.5	22.5	18.3	15.8	12.5
		MCDB	11.6	9.8	14.7	17.9	24.9	29.4	28.8	28.6	28.7	20.2	16.5	13.0
	2%	WB	7.4	5.1	7.8	12.0	16.5	20.8	22.4	22.2	20.7	16.5	13.5	9.1
		MCDB	8.0	6.2	9.3	14.6	21.7	26.2	27.7	26.9	24.6	18.4	14.4	9.5
	5%	WB	3.6	2.9	5.4	9.9	14.8	19.3	21.4	21.1	19.2	15.1	11.3	6.2
		MCDB	4.4	4.2	7.4	12.9	19.1	23.9	26.4	25.8	22.2	17.0	12.3	6.9
	10%	WB	0.9	1.2	3.5	8.1	13.2	17.8	20.4	19.9	17.9	13.2	8.9	3.4
		MCDB	1.7	2.4	5.6	11.4	17.0	22.2	24.7	23.8	20.7	15.1	10.2	4.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.4	10.5	10.9	12.2	12.1	12.1	12.3	12.5	11.1	9.1	9.4
		MCDBR	13.6	12.1	13.2	16.2	17.9	16.8	15.1	15.0	15.2	14.2	11.7	11.7
		MCWBR	13.2	10.2	9.4	9.9	10.1	9.2	7.7	7.8	9.0	9.8	10.0	11.1
	5% WB	MCDBR	13.7	10.9	11.6	12.9	14.6	14.3	12.8	12.7	12.3	11.8	10.2	11.2
		MCWBR	13.5	9.9	9.5	9.4	9.1	8.6	7.2	7.2	7.8	9.3	9.8	11.3
Clear-Sky Solar Irradiance	taub	0.279	0.291	0.320	0.348	0.369	0.408	0.425	0.397	0.362	0.328	0.308	0.291	
	taud	2.478	2.447	2.455	2.429	2.414	2.319	2.288	2.368	2.436	2.534	2.545	2.489	
	Ebn at Noon	847	902	916	913	900	862	841	853	858	847	806	791	
	Edh at Noon	68	84	96	108	113	125	128	114	98	77	64	61	
All-Sky Solar Radiation	RadAvg	1.51	2.38	3.54	4.38	5.00	5.43	5.57	4.95	4.04	2.55	1.51	1.16	
	RadStd	0.14	0.20	0.24	0.41	0.60	0.45	0.43	0.31	0.28	0.24	0.16	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	+82	N/A

Nomenclature: See separate page